Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- **Rows**: 3,900
- **Columns**: 18
- **Key Features**:
    - Customer demographics *(Age, Gender, Location, Subscription Status)*
    - Purchase details *(Item Purchased, Category, Purchase Amount, Season, Size, Color)*
    - Shopping behavior *(Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)*
- **Missing Data**: *37 values in Review Rating column*

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.

- **Initial Exploration:** Used `df.info()` to check structure and `df.describe()` for summary statistics.

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

```
...   <class 'pandas.core.frame.DataFrame'>
      RangeIndex: 3900 entries, 0 to 3899
      Data columns (total 18 columns):
       #   Column                    Non-Null Count  Dtype
      ---  ------                    --------------  -----
       0   Customer ID               3900 non-null   int64
       1   Age                       3900 non-null   int64
       2   Gender                    3900 non-null   object
       3   Item Purchased            3900 non-null   object
       4   Category                  3900 non-null   object
       5   Purchase Amount (USD)     3900 non-null   int64
       6   Location                  3900 non-null   object
       7   Size                      3900 non-null   object
       8   Color                     3900 non-null   object
       9   Season                    3900 non-null   object
       10  Review Rating             3863 non-null   float64
       11  Subscription Status       3900 non-null   object
       12  Shipping Type             3900 non-null   object
       13  Discount Applied          3900 non-null   object
       14  Promo Code Used           3900 non-null   object
       15  Previous Purchases        3900 non-null   int64
       16  Payment Method            3900 non-null   object
       17  Frequency of Purchases    3900 non-null   object
      dtypes: float64(1), int64(4), object(13)
      memory usage: 548.6+ KB
```

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
    - Created **age_group** column by binning customer ages.
    - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script Duckdb SQL Query library and registered the cleaned DataFrame into the table for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in SQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
0	Male	157890.0
1	Female	75191.0

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
0	2	64
1	3	73
2	4	90
3	7	85
4	9	97
...	...	...
834	1667	64
835	1671	73
836	1673	73
837	1674	62
838	1676	90

839 rows × 2 columns

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
0	Gloves	3.86
1	Sandals	3.84
2	Boots	3.82
3	Hat	3.80
4	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	avg_pur_amt
0	Standard	58.46
1	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_customers	avg_spend	total_revenue
0	No	2847	59.87	170436.0
1	Yes	1053	59.49	62645.0

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
0	Hat	50.00
1	Sneakers	49.66
2	Coat	49.07
3	Sweater	48.17
4	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	no_of_customers
0	Returning	701
1	Loyal	3116
2	New	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
0	1	Footwear	Sandals	160
1	2	Footwear	Shoes	150
2	3	Footwear	Sneakers	145
3	1	Accessories	Jewelry	171
4	2	Accessories	Sunglasses	161
5	3	Accessories	Belt	161
6	1	Outerwear	Jacket	163
7	2	Outerwear	Coat	161
8	1	Clothing	Pants	171
9	2	Clothing	Blouse	171
10	3	Clothing	Shirt	169

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

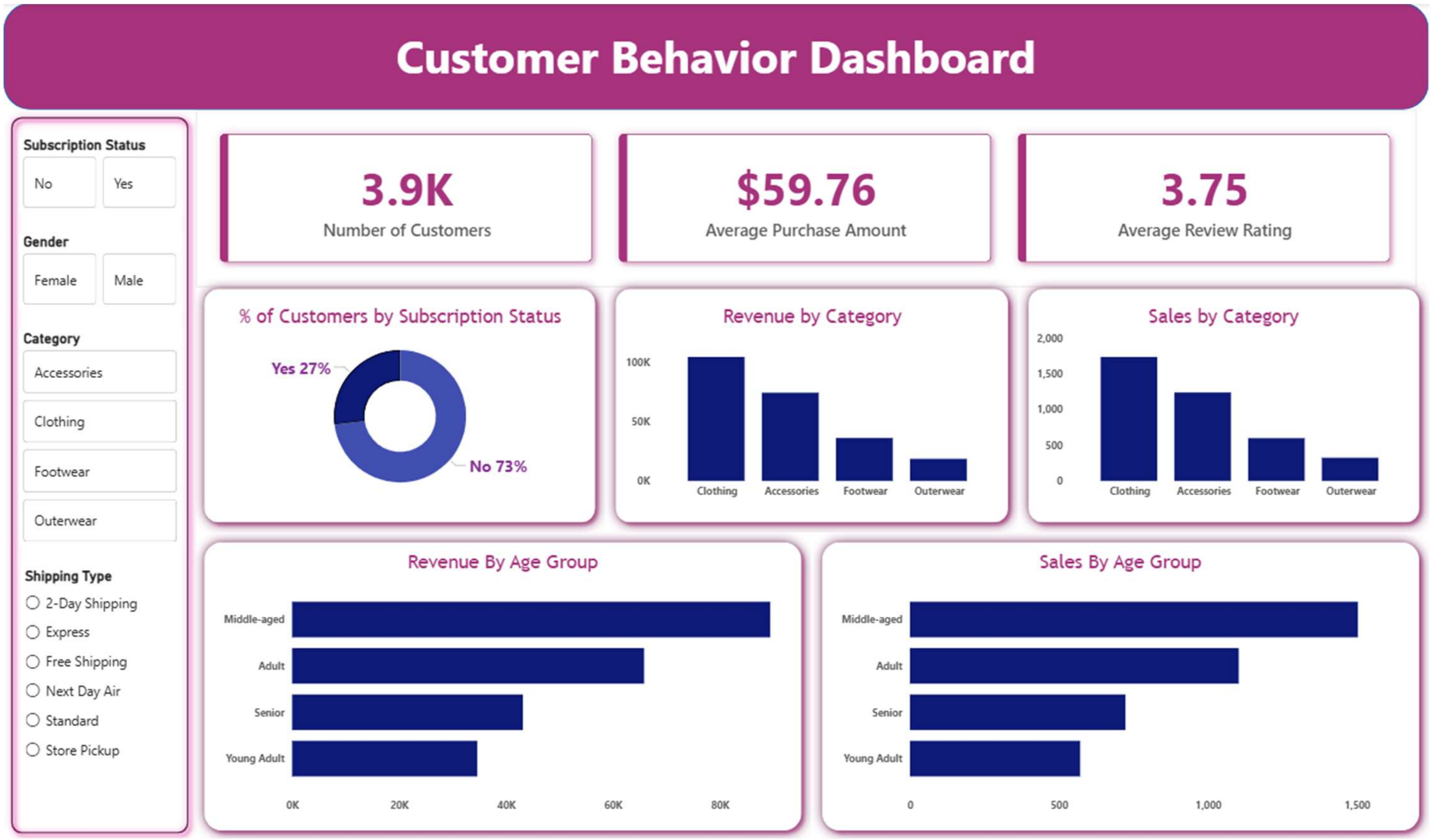
	subscription_status	repeat_buyers
0	Yes	958
1	No	2518

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group	total_customers	avg_pur_amt	total_revenue
0	Middle-aged	1503	59.51	89445.0
1	Adult	1103	59.69	65842.0
2	Senior	723	59.70	43164.0
3	Young Adult	571	60.65	34630.0

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.